Layer 7 Load Balancers

Linux has proven itself as a rock-solid operating system platform for industry-leading software appliances and applications, one of which is for load-balancing. As global Internet traffic increases, it demands an increased throughput from the existing infrastructure. It is crucial to deliver content fast; this is especially true for businesses whose only interface with clients is their Web portals. Load balancers add great value in this case, and also provide multiple other functionalities. This article explains new trends in this well-known product category, which are not adequately explored by IT managers and systems administrators.

Why is there a need for load balancers? While managing a Web services infrastructure, Web administrators often find it a challenge to cope with increased website hits, while maintaining high availability of the servers. This situation gets even tougher when a new Web application or functionality is released, attracting more users per day. Optimisation of server performance is thus a continuous job.

Consider a Web server hosting a site running a few applications. When the site gains more users, there are many more page requests. Serving each request uses a definite amount of CPU, memory and network resources. Adding powerful resources can only solve the problem to some extent, while introducing another challenge. When the Web server hits the ceiling in terms of its resource limit, it starts dropping Web requests, which results in a bad user experience—a 'broken' or 'hanging' Web page. And if the Web server goes down, for some reason, the entire site becomes non-functional. This can certainly result in a loss of reputation, and in some cases, also a monetary loss for the organisation. To pre-empt such situations, IT management teams must deploy load-balancing solutions in the data-centre infrastructure. We will soon discuss how a load balancer can not only distribute traffic, but also help ease network operations tasks.

How does a load balancer work?

First-generation balancing devices were implemented around BSD UNIX versions. A new trend of balancing products is typically in the form of an appliance running a Linux distribution; some enterprise-grade appliances use Red Hat or similar Linux flavours. Functionally, a load balancer can balance traffic by distributing it among two or more servers. Figure 1 shows a typical Web farm configuration with a load-balancing device that acts as a front-end server to handle all Web requests. Each silo hosts a different set of applications, whereas all servers in a given silo host identical applications. From the configuration point of view, the device is configured with two separate IP ranges. One is used to handle incoming traffic, and the other, called virtual servers, is used to connect to the nodes under its control. Thus, it forms an agent service between the requesting client and the responding server. It also acts on the requests intelligently, based on the configured rules, to choose a recipient node with the least work-load at that particular time.

Rules define how a request should be handled, and also how to handle special conditions such as node preference, session management, etc. The load-balancing device then makes a separate TCP connection with the recipient Web server, and redirects the requests to it, while it keeps track of the request processing.